

Functional Pin Diagram & Signals of 8086

Q. Discuss the functional pin diagram of the 8086 microprocessor. Explain the significance of the multiplexed Address/Data bus and the pins specific to Minimum and Maximum modes.

[DEF] Definition to Remember

The Intel 8086 is a 40-pin IC. To save pin count, it uses multiplexed buses (pins share dual roles based on the clock cycle). A unique feature is the MN/MX (Pin 33)' pin, which toggles the processor between Minimum Mode (single-processor) and Maximum Mode (multi-processor).

1. Multiplexed Address/Data and Status Buses

* AD0 - AD15 (Pins 16-2, 39): Multiplexed Address/Data. During clock cycle T1, they carry the lower 16 bits of the memory address. During T2-T4, they carry 16-bit data.

2. Common Control & Interrupt Pins

* ALE (Address Latch Enable): Signals external latches to capture the address from the AD bus during T1.

3. Minimum Mode Pins (MN/MX (Pin 33)' tied to +5V)

Operates as a standalone processor generating its own control signals.

4. Maximum Mode Pins (MN/MX (Pin 33)' tied to GND)

Used in multi-processor systems (e.g., with 8087 Math Coprocessor). Control signal generation is handed to an external 8288 Bus Controller.

[MUST-WRITE] Must-Write Points (for 10 marks)

1. The 8086 is a 40-pin IC utilizing multiplexed buses to save pins.
2. AD0-AD15: Act as Address during T1, and Data during T2-T4.
3. ALE (Address Latch Enable): Tells external latches to grab the address during T1.
4. Can operate in Minimum Mode (standalone) or Maximum Mode (multi-processor) using the MN/MX (Pin 33)' pin.
5. Minimum Mode: CPU generates its own signals (M/IO', WR', HOLD/HLDA).
6. Maximum Mode: Control is handed to an 8288 Bus Controller. Uses S0, S1, S2 to tell 8288 what to do.
7. Coprocessors use RQ'/GT' (Request/Grant) to take the bus in Maximum mode.

[RECALL] Quick Recall

40-Pin IC -> Multiplexed (AD0-AD15: Addr T1, Data T2-T4) -> ALE (Latches Addr) -> Min Mode (Standalone, Generates WR, M/IO, HOLD)
-> Max Mode (Multi-processor, Uses 8288 Controller, S0-S2, RQ/GT)